Curriculum Vitae

Sina Dadras

Address: Tehran, Iran
Phone Number: +98-9122144790
Email: sinadadras021@gmail.com
LinkedIn: www.linkedin.com/in/sina-dadras-86148a2a2
GitHub: github.com/sinadadras021

EDUCATION

M.Sc. in Environmental Engineering (Air Pollution) — (2023 - 2025)

University of Tehran, Department of Environmental Engineering, Faculty of Environment — Tehran, Iran

GPA: 18.44/20 (US equivalent: 4.0/4.0) (Ranked 1st in the program; Equivalent to UK First-Class Honors)

Thesis: "Hourly Ozone Prediction Using Ensemble Forecasting to Accurately Capture Peak Concentration: A Case Study of Tehran, Iran"
Advisors: Prof. Khosro Ashrafi (khashrafi@ut.ac.ir), Asst Prof. Majid Shafiepour Motlagh (shafiepour@ut.ac.ir)

Selected Courses (all A):

- Environmental Engineering Fundamentals: **19.25/20**
- Air Pollution Modeling: **19/20**
- New Methods of Measuring Air Pollutants: **19.75/20**
- Geographic Information and Remote Sensing in Air Pollution: **19/20**
- Application of Nano and Biotechnology in Air Pollution Control: **19/20**
- Control of Gaseous Emission Pollutants from Stationary Sources: **19.75/20**

B.Sc. in Civil Engineering — (2016 - 2021)

Iran University of Science and Technology (IUST), School of Civil Engineering — Tehran, Iran

GPA: 13.64/20 (Equivalent to UK 2:1 Upper Second-Class Honours)

RESEARCH INTERESTS

Air Pollution Modeling & Air Quality Prediction, AI Interpretability/mechanistic understanding, Atmospheric Chemistry, Remote Sensing, Integration of Physics-informed Machine Learning and Advanced Deep Learning techniques for Complex Data Analysis.

RESEARCH EXPERIENCE

Air Pollution Forecasting Using Deep Learning and Ensemble Methods (2025-2026)

University of Tehran — Advisors: Prof. Khosro Ashrafi & Asst. Prof. Majid Shafiepour Motlagh

- **Hourly Ozone Forecasting:** Developed the Residual-Aware Meta Ensemble (RAME), a hybrid deep learning framework combining bidirectional recurrent units and dilated convolutions for 1–24-hour ozone concentration forecasting in Tehran. Achieved city-wide mean 24-hour $R^2 = 0.87$, RMSE = 8.12 ppb, and MAE = 5.30 ppb, accurately identifying 96% of critical ozone days.

- **Daily-Averaged $PM_{2.5}$ Prediction:** Designed next-day and day-after-next $PM_{2.5}$ daily average forecasting models based on ensemble learning and meta-learning strategies. The MLP-Meta model achieved RMSE = 7.15 µg/m³ and IOA = 0.914 on Day 1, and RMSE = 8.21 µg/m³ and IOA = 0.872 on Day 2, while the early warning system detected 78.0% of critical episodes with only 2.5% false alarms.

Reproducible Research Software: Released both forecasting frameworks as documented, tested open-source packages, each mapping every equation and table of the corresponding manuscript to the function implementing it. Data-leakage controls are enforced at runtime rather than by convention, and the test suites pin each default to the published method so that code and manuscript cannot silently diverge. Both run end-to-end from raw monitoring and meteorological input to forecast output in a single command.

PUBLICATIONS

1. Dadras, S., Hajighasemi, E., Ashrafi, K., & Shafiepour Motlagh, M. (2026).
   Hourly Ozone Prediction Using Ensemble Forecasting to Accurately Capture Peak Concentration: A Case Study of Tehran, Iran.
   Environmental Science and Pollution Research.
   https://doi.org/10.1007/s11356-026-38197-7
   Code: github.com/sinadadras021/O3-RAME-forecasting

2. Hajighasemi, E., Dadras, S., Ashrafi, K., & Shafiepour Motlagh, M. (2026).
   Systematic Comparison of Ensemble Learning Strategies for Multi-Day $PM_{2.5}$ Forecasting in Urban Environments.
   Process Safety and Environmental Protection (Under Review).
   Code: github.com/sinadadras021/PM25-Ensemble-forecasting

HONORS and AWARDS

- **Ranked 1st** in M.Sc. program, Environmental Engineering (Air Pollution), University of Tehran.
- **62nd** of 16,000+ candidates — Civil Engineering Graduate Admission Exam (National).
- **Top 1%** of 150,000+ candidates — National Entrance Exam for Bachelor's Degree.

PROFESSIONAL EXPERIENCE

Air Pollutant Dispersion Modeling — Course Project, University of Tehran, Tehran, Iran (Academic Project)
Developed a Python Gaussian plume dispersion model for multiple point sources — stability classification, plume rise, wind-profile and mixing-layer effects, dry deposition and first-order chemical decay. Later rebuilt as an open-source package with a graphical interface, verified against the closed-form Gaussian solution to machine precision and against conservation of mass, and released with a ten-figure validation gallery spanning all six stability classes (github.com/sinadadras021/gaussian-plume-model).

Teaching Assistant, Air Pollution Modeling Course — University of Tehran, Tehran, Iran (Academic Appointment)
Assisted in teaching graduate students the fundamentals of air pollution modeling and Python-based implementation. Developed and delivered tutorials on data-driven modeling techniques, data

analysis, and visualization, and provided one-on-one support to strengthen students' understanding of atmospheric dispersion concepts and computational tools.

LANGUAGE and TECHNICAL SKILLS

Languages

- English (C1) — IELTS Overall Band 7.0 (CEFR Level C1) (September 2025)

Listening: 7, Reading: 7, Writing: 6, Speaking: 7

- Persian: Native

Technical

- Programming & Data: Python (NumPy, Pandas, scikit-learn), TensorFlow, PyTorch; data analysis and visualization; GIS (ArcGIS).
- Scientific Software Practice: Reproducible research code (Git, packaging, configuration-driven pipelines); automated testing against analytical solutions and conservation laws; technical documentation for external readers.
- Air Quality & Modeling: Gaussian plume dispersion modeling; urban-scale ozone and $PM_{2.5}$ forecasting; time-series modeling and validation.
- Quantitative Methods: Statistics for environmental data; calculus and applied mathematical modeling.

REFERENCES

Prof. **Khosro Ashrafi** — M.Sc. Thesis Advisor, University of Tehran, Faculty of Environment (khashrafi@ut.ac.ir)

Asst. Prof. **Majid Shafiepour Motlagh** — M.Sc. Thesis Advisor, University of Tehran, Faculty of Environment (shafiepour@ut.ac.ir)

Asst. Prof. **Mohsen Mir Mohammadi** — Course Instructor, University of Tehran, Faculty of Environment (mirmohammadi.m@ut.ac.ir)